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Comparison of different criteria in the evaluation of molecular targeted therapies in patients with hepatocellular carcinoma: a meta-analysis.

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-052294
Article Type:	Original research
Date Submitted by the Author:	12-Apr-2021
Complete List of Authors:	Bai, Yuping; shan dong sheng li yi yuan, Department of Gastroenterology Xie, Xiaoyu; shan dong sheng li yi yuan, Department of Gastroenterology Feng, Yuemin; shan dong sheng li yi yuan, Department of Gastroenterology Yang, Yao; shan dong sheng li yi yuan, Department of Gastroenterology Zhu, Qiang; shan dong sheng li yi yuan, Department of Gastroenterology
Keywords:	Hepatobiliary disease < GASTROENTEROLOGY, Hepatology < INTERNAL MEDICINE, Hepatobiliary tumours < ONCOLOGY

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Comparison of different criteria in the evaluation of molecular targeted therapies in patients with hepatocellular carcinoma: a meta-analysis.

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Conflicts of Interest and Source of Funding:

None of the authors have identified a conflict of interest. This work was funded by the National Natural Science Foundation of China (No. 81770607 and 81772626).

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Abstract

Objective Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST 1.1) and modified RECIST (mRECIST) are commonly used to assess tumor response. Which one is better to evaluate efficacy for patients with hepatocellular carcinoma (HCC) after molecular targeted therapies is still controversial. We conducted this meta-analysis to compare the tumor response and correlation with overall survival (OS) between the two criteria.

Design Meta-analysis.

Data sources EMBASE, PubMed, Web of Science and Cochrane Library were used to search the studies from inception to 30 June 2020.

Methods A comprehensive search was performed in databases EMBASE, PubMed, Web of Science and Cochrane Library. Inter-method agreement between the two criteria was measured using the k coefficient. Hazard ratio (HR) for overall survival was used for meta-analysis, considering responders versus non-responders according to RECIST1.1 and mRECIST criteria.

Results 19 studies comprising 1,432 patients were included. The agreement of tumor response between the two criteria was moderate for objective response (k=0.469, 95% CI 0.328-0.576) and perfect for progressive disease (k=0.900, 95% CI 0.851-0.944). The overall response rate according to mRECIST was significantly higher than

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RECIST1.1 (22.9% versus 10.4%, p<0.001). The disease control rate was similar according to mRECIST and RECIST1.1 (68.4% versus 66.9%, p=0.5). OS was significantly longer in patients with response than patients with non-response according to mRECIST (HR=0.56, 95% CI=0.41-0.78, p=0.0004), however the RECIST1.1 couldn't distinguish the OS of the responders and the non-responders (HR=0.68, 95%CI, 0.44-1.05, p=0.08).

Conclusion The mRECIST appears more accurate than RECIST1.1 to evaluate the efficacy of molecular targeted therapies in patients with HCC.

PROSPERO registration number CRD42020200895.

Ethics approval Ethics approval is not required in this meta-analysis.

Key Words:

Hepatocellular carcinoma; Molecular targeted therapies; Modified Response Evaluation Criteria in Solid Tumors; Response Evaluation Criteria in Solid Tumors version 1.1; Meta-analysis

Strengths and limitations of this study

1. This meta-analysis is the first comprehensive study to compare the two criteria in assessing tumor response of molecular targeted therapies in patients with hepatocellular carcinoma.

2. Our results showed the results were reliable, there were no significant heterogeneity among the studies and no significant publication bias.

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3. Because of the limited number of studies, we could not compare the two criteria in the subgroup with the different kinds of therapeutic agents and different time interval between follow-up imaging examinations.

Abbreviations:

BCLC, Barcelona Clinic Liver Cancer; CI, confidence interval; CR, complete response; CT, computed tomography; DCR, disease control rate; EASL, European Association for the Study of the Liver; ECOG PS, Eastern Cooperative Oncology Group performance status; ESMO, European Society for Medical Oncology; HCC, hepatocellular carcinoma; HR, hazard ratio; mRECIST, modified Response Evaluation Criteria in Solid Tumors; MRI, magnetic resonance imaging; NOS, Newcastle-Ottawa scale; OR, objective response; ORR, overall response rate; OS, overall survival; PD, progressive disease; PET, positron emission tomography; PR, partial response; PRISMA, Preferred Reporting Items for Systematic reviews and Meta-Analyses; RECIST 1.0, Response Evaluation Criteria in Solid Tumors version 1.0; RECIST 1.1, Response Evaluation Criteria in Solid Tumors version 1.1; SD, stable disease; WHO, World Health Organization.

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Introduction

Hepatocellular carcinoma (HCC) is the sixth most common cancer and the fourth leading cause of cancer-related mortality worldwide [1]. There have been significant advances in treatment for HCC over the past ten years. Available therapeutic options contain surgical resection, liver transplantation, ablative techniques, transarterial chemoembolization, transarterial radioembolization, radiotherapy and molecular targeted therapies [2]. Molecular targeted therapies are indicated for patients with advanced tumors or earlier stage tumors unsuitable for surgical resection or locoregional therapies [3]. It has already been confirmed that molecular targeted therapies can improve survival in patients with HCC due to their unique antiproliferative and antiangiogenic function [4]. The accurate assessment of therapeutic efficacy of molecular targeted therapies is essential for routine anti-cancer treatment as well as clinical trials.

Radiological evaluation of tumor response is a well-recognized surrogate endpoint in the assessment of therapeutic efficacy of molecular targeted therapies in patients with HCC [5], which is crucial to help identify potentially resistant patients, avoiding unnecessary toxicities. Tumor response was initially measured according to the World Health Organization (WHO) criteria and Response Evaluation Criteria in Solid Tumors version 1.0 (RECIST 1.0) guideline [6 7]. However, they have been shown poor correlation with survival outcome in patients with HCC [8-10]. Nowadays, the RECIST 1.1 and modified RECIST (mRECIST) criteria are the most commonly used criteria to assess tumor response. The major changes in RECIST 1.1 include the reduction in the

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number of target lesions and the augmented definition of disease progression [11], which relies on the change in the sum of the greatest diameters. The mRECIST has been developed which differ from RECIST 1.1 in that the target lesion measured is not the whole lesion but only the viable tumor, defined as the contrast-enhanced portion of the tumor on hepatic arterial phase images[8].

European Association for the Study of the Liver (EASL) and European Society for Medical Oncology guidelines (ESMO) suggested applying mRECIST or RECIST 1.1 in patients with HCC treated with molecular targeted therapies [3] [12]. However, National Comprehensive Cancer Network guideline indicated that validated criterion to evaluate tumor response to molecular targeted therapies between the two criteria is needed [13]. Besides, several studies demonstrated that overall survival can be predicted more accurately by mRECIST than RECIST 1.1, since the latter is not capable of assessing therapy induced intratumoral necrosis [14] [15]. On the contrary, another study observed both methods provided correlation with OS equally[16]. Which set of criteria is better to assess response to molecular targeted therapies remains controversial.

We conducted this meta-analysis to compare the tumor response of patients with HCC after molecular targeted therapies by RECIST1.1 and mRECIST and determine which criterion correlates better with overall survival.

Material and methods

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Protocol and guidance

This study was conducted according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) and registered in PROSPERO (CRD42020200895).

Patient and public involvement

Patients and the public were not involved in this meta-analysis.

Search strategy

A comprehensive search of the literature was performed in electronic databases EMBASE, PubMed, Web of Science and Cochrane Library up to 30 June 2020. The following Mesh terms and text words were confined to the title or abstract: “RECIST”, “mRECIST”, “Response Evaluation Criteria in Solid Tumors”, “liver cancer” and “hepatocellular carcinoma”. In addition, we surveyed all the references of relevant articles.

Selection of studies

To be eligible for inclusion, studies had to meet the following criteria: (1) the diagnosis of HCC was based on pathology or radiological findings, in accordance with the criteria of practice guidelines; (2) patients with HCC must be treated with molecular targeted therapies; (3) response assessment after molecular targeted therapies was evaluated according to both RECIST 1.1 and mRECIST criteria; (4) available data about overall

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survival and k coefficient or sufficient information to calculate it. General exclusion criteria were: (1) presence of an additional primary malignancy in other organ; (2) patients with HCC received other therapies; (3) case analysis, letters, reviews and expert opinions; (4) studies with incomplete data.

Quality assessment

The Newcastle-Ottawa scale (NOS) was used to assess the quality of the studies, this scale consists of three factors: the selection of patients, comparability of the study groups, and assessment of outcome[17]. The maximum total score on this scale is 9 and studies with scores ≥6 were defined as high-quality studies.

Data extraction

Data from the included studies were summarized independently by two of the authors, who were blinded to the journals, authors and institutions of all available articles. By reading the full texts of the selected studies, two investigators extracted the following data: name of all authors, year of publication, number of enrolled patients, age, sex, ECOG PS, Child-Pugh score, BCLC stage, tumor number, tumor size, type of treatment, reported hazard ratio (HR) for overall survival (OS) according to mRECIST and RECIST 1.1 criteria and k coefficient of concordance in each study.

Tumor response assessment

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Evaluation of tumor response according to the RECIST 1.1 was defined as follows: complete response (CR) is the disappearance of all target lesions; partial response (PR) is at least a 30% decrease in the sum of the diameters of the target lesions; progressive disease (PD) is at least a 20% increase in the sum of the longest diameters of target lesions or the appearance of one or more new lesions; stable disease (SD) is neither sufficient shrinkage to qualify for PR nor sufficient increase to qualify for PD. According to mRECIST, CR is defined as the disappearance of any intratumoral arterial enhancement in all target lesions; PR is at least a 30% decrease in the sum of diameters of viable target lesions, taking as reference the baseline sum of the diameters of target lesions; PD is an increase of at least 20% in the sum of the diameters of viable target lesions, taking as reference the smallest sum of the diameters of viable target lesions recorded since the treatment started; SD is any cases that do not qualify for either PR or PD. Objective response (OR) included both CR and PR, and disease control included CR, PR and SD.

Statistical analysis

Inter-method agreement between similar categorical items of the two criteria was measured using the k coefficient. The agreement was interpreted as poor ($k < 0$), slight ($k = 0$ to 0.20), fair ($k = 0.21$ to 0.40), moderate ($k = 0.41$ to 0.60), substantial ($k = 0.61$ to 0.80), and almost perfect ($k > 0.80$)[18]. The overall response rate (ORR) and disease control rate (DCR) between the two criteria were compared by the chi-square tests with the significance at $p<0.05$. Hazard ratio (HR) for overall survival was used for meta-

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analysis, considering responders (complete or partial response) versus non-responders, using the generic inverse variance outcome type in Cochrane Collaboration's Review Manager 5.4. The heterogeneity was quantified using the I^2 statistic. A fixed-effects model was used unless there was obvious heterogeneity among the included studies, funnel plots and Egger's test were used to grossly exclude publication bias.

Results

Eligible studies for analysis

A total of 5,922 studies were retrieved from the electronic database search. After removing duplicates, 4,126 titles and abstracts were further examined. 71 publications underwent full text review to determine their eligibility for the meta-analysis and 52 were excluded. Finally, 19 studies including 1,432 patients comparing tumor response between the RECIST 1.1 and mRECIST criteria were included [14-16 19-34] (**Figure 1**).

13 studies were excluded because they did not have comparison group, which only focused on mRECIST or RECIST 1.1 criteria. One research was written with Chinese. 11 studies compared with other methods, such as Choi and EASL criteria et al. 14 studies were excluded because they used other imaging examinations or other evaluative methods, such as RECIST1.0, positron emission tomography (PET) imaging and contrast-enhanced ultrasonography. Six studies were excluded because they were all conference abstracts with incomplete data. Three studies were excluded for lack of original data and had insufficient data for extraction. Three studies were excluded for

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unavailable for full text. One study focused on investigating the tumor response using manual and automatically extracted measurements.

Summary of baseline characteristics

A total of 1,432 patients with HCC treated with molecular targeted therapies were collected from the 19 studies (**Table 1**). 724 patients treated with sorafenib [14-16 19-25 27 34], 30 patients treated with axitinib [26], 289 patients treated with lenvatinib [28 30 31 33], 379 patients treated with regorafenib [29] and 10 patients treated with ramucirumab [32]. Most studies included Child-Pugh class A and a minority of Child-Pugh class B patients, only two studies included a small percentage of Child-Pugh class C patients [27 29].

Evaluation of tumor response was performed according to the RECIST 1.1 and mRECIST criteria and assessment of response was carried out by contrast-enhanced spiral computed tomography (CT) or gadolinium-enhanced magnetic resonance imaging (MRI) after 4–8 weeks from treatment, depending on each study (**Table 2**).

Risk of bias within studies

All of 19 studies had good quality. The quality of included studies assessed by NOS was 6 to 8.

Comparison of tumor response between the RECIST 1.1 and mRECIST criteria.

The agreement and disagreement of tumor response of the two criteria were described in **Table 3**, which could be available or calculated from 7 studies including 361 patients

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[15 16 20-22 25 26]. For objective response, the agreement of tumor response between the two criteria was moderate (k=0.469, 95% CI 0.328-0.576). Of 213 patients with SD according to RECIST1.1, 42 patients were reclassified to overall response according to mRECIST. For disease control, the agreement of tumor response between the two criteria was almost perfect (k=0.900, 95% CI 0.851-0.944). Of 116 patients with PD according to mRECIST, only 5 patients were reclassified to SD according to RECIST1.1.

Of the included 1,432 patients with HCC, 1,365 patients were evaluated according to both RECIST1.1 and mRECIST criteria. 33 patients treated with sorafenib, 9 patients treated with axitinib and 25 patients treated with lenvatinib were not unevaluable, so we excluded these patients. **Table 4** shows the tumor response assessed by RECIST 1.1 and mRECIST after molecular targeted therapies in the 19 considered studies. The ORR according to mRECIST was significantly higher than RECIST1.1 (22.9% versus 10.4%, p<0.001). For disease control rate (DCR), one study was considered not eligible for its incomplete date [29]. The DCR was similar according to mRECIST and RECIST1.1 (68.4% versus 66.9%, p=0.5).

Survival analysis according to the RECIST 1.1 and mRECIST criteria.

Of the nineteen articles, 12 studies were excluded for their incomplete data. Finally, seven studies were used to produce survival analysis [14 15 21 23 27 30 34]. The ORR according to RECIST 1.1 and mRECIST criteria was 8.25% and 18.73%, respectively. OS was

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significantly longer in patients with response than patients with non-response according to mRECIST (HR=0.56, 95% CI=0.41-0.78, p=0.0004) (**Figure 2**), with no significant heterogeneity among the studies (I^2=0, p=0.93). However, the RECIST1.1 could not distinguish the OS of the responders and the non-responders (HR=0.68, 95%CI, 0.44-1.05, p=0.08) (**Figure 3**), with no significant heterogeneity among the studies ((I^2=0, p=0.43). With the limitation of the small number of studies, funnel plots for both RECIST 1.1 and mRECIST did not show asymmetry (**Supplementary figure 1-2**). Egger's test also showed there was no clear evidence of publication bias (p=0.052 for RECIST1.1 and p=0.503 for mRECIST).

Discussion

Tumor response assessment is very important in the management of cancer, both in clinical practice and it can be a surrogate end point of survival for evaluation of therapeutic efficacy in clinical studies [35 36]. In particular, with the increasing use of molecular targeted therapies in patients with HCC.

Since new molecular targeted agents exert their antitumoral activity by inducing tumor necrosis, with rare changes in volume shrinkage, traditional WHO and RECIST criteria do not always represent an appropriate tool for response evaluation. Anatomic imaging alone may have limitations, particularly in assessing the activity of targeted therapies which stabilize diseases. This promoted the development of the mRECIST for a

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response that incorporated treatment-induced tumor necrosis by dynamic imaging. In this study, we compared the role of the RECIST 1.1 and mRECIST criteria in the evaluation of molecular targeted therapies in patient with HCC.

We investigated the concordance between the RECIST 1.1 and mRECIST criteria for the assessment of tumor response in patients with HCC treated with molecular targeted therapies. Our results showed that there was a considerable discrepancy in the assessment of tumor responses between the RECIST 1.1 and mRECIST criteria. When adopting the mRECIST, the ORR was significantly increased, suggesting significant clinical impact of mRECIST on making therapeutic decision. There are several possible reasons. Firstly, molecular targeted therapies are based on the inhibition of a number of pro-angiogenic signaling pathways, which stimulating angiogenesis, are responsible of the characteristic hyper-vascular pattern of HCC lesions [37]. The therapeutic response to them correlates with changes in tumor structure, including decreased vascularization and increased tissue necrosis or cavitation, but it is not always associated with reduction in tumor size [38]. Secondly, HCC and cirrhosis coexist in more than 80% of cases creating unique complexities for imaging assessment. Pathogenic and hemodynamic changes inherent to cirrhosis may mimic or conceal intrahepatic tumors [39].

We also fund that there was an excellent agreement in the assessment of the disease progression between the RECIST 1.1 and mRECIST criteria, both of the criteria are equally able to discriminate progressors and non-progressors, which is the most

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relevant parameter in clinical practice. Disease progression seems to involve an increase in vascularization that immediately translates into an increase in the greatest lesion dimension.

Besides, our results showed mRECIST can be of help in predicting long-term survival in patients treated with molecular targeted therapies and that those patients with an OR had significantly better survival compared to patients who only achieve SD or PD. However, the OS of those classified as respondent by RECIST1.1 did not differ significantly from that of non-responders. Edeline et al demonstrated that in the 79.2% of patients classified as stable by RECIST 1.1, the use of mRECIST enabled the prediction of different prognostic subgroups with a significantly better median OS of 17.1 months for responding patients compared with 9.7 months for stable patients [16]. Our results suggest that mRECIST may offer a suitable alternative to RECIST in Phase II clinical trials, in which detection of an efficacy signal is paramount.

We also found that most of included patients were classified as PD. Patients with PD always have a poor outcome. Treatment beyond radiological progression is not warranted and that patients should be actively monitored for radiological progression rather than waiting for symptomatic progression. A recent review indicated that 'Progressive disease' concept includes different patterns of progression characterized by different prognosis. The reason for the imperfect correlation between surrogate end point and OS likely relies on the basis that not all patterns of progressions are equal in

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terms of prognostic implications. In a brilliant paper, Reig et al. demonstrated that the appearance of new extrahepatic lesions has a far worse prognostic impact than the enlargement of pre-existing lesions or the appearance of new intrahepatic nodules [40]. Thus, a careful evaluation of the progression pattern is indeed required in clinical practice before switching to a second line treatment.

There are several limitations in our study. First, most of the included studies were retrospective. Second, this study included heterogeneous patients with different kinds of therapeutic agents and the variable interval between follow-up imaging examinations. In addition, because of the limited number of studies, we could not compare two criteria in the subgroup with the same drug. It is necessary to verify these results in studies with larger homogeneous patients' cohort.

In conclusion, mRECIST appears more appropriate than RECIST 1.1 to identify responders with long survival benefiting from molecular targeted therapies in patients with HCC, and RECIST 1.1 can be used for atypical lesions. This alternative criterion warrants further evaluation in addition to RECIST 1.1 in prospective trials exploring antiangiogenic drugs in advanced HCC.

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Acknowledgements

This work was funded by the National Natural Science Foundation of China (No. 81770607 and 81772626).

Specific author contributions: Yuping Bai contributed to to the design of the study, data collection, data analysis and article writing; Xiaoyu Xie contributed to data collection; Yuemin Feng and Yao Yang contributed equally to the data analysis. Qiang Zhu and Ping Yao contributed to the conception and design of the study and critical revision. All authors contributed to the revision and gave final approval of the article.

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Table 1. Baseline characteristics of include studies

Study	Year	Country	Patients, N (n) †	Treatment	Women, No. (%)	Age, y	ECOG PS	Child-Pugh class	BCLC stage	Tumor number	Tumor size (mm)
Spira et al	2011	Germany	25	Sorafenib	2 (8.0%)	65 (42–75) ‡	0: 6 1 or 2: 19	A: 23 B: 2	NR	NR	NR
Murakami et al	2011	Japan	27	Sorafenib	1 (3.7%)	63	NR	A: 23 B: 4	NR	NR	NR
Edeline et al	2012	France	53	Sorafenib	5 (9.4%)	NR	0: 29 1: 24	A: 41 B: 12	B: 22 C: 31	NR	NR
Moschouris et al	2012	UK	21	Sorafenib	2 (9.5%)	66.8±8.5 §	NR	A: 10 B: 11	B: 9 C: 12	Solitary: 9 Multifocal: 12	NR
Kawaoka et al	2012	Japan	66 (49)	Sorafenib	8 (12.1%)	63 (35-80) ‡	NR	A: 58 B: 8	0: 2 A: 1 B: 14 C: 49	1–2: 16 />3: 50	50 (8.3–194) ‡
Arizumi et al	2014	Japan	156	Sorafenib	36 (23.1%)	73 (66-78) ‡	0: 150 1: 5 2: 1	A: 129 B: 27	A: 39 B: 36 C: 81	NR	NR
Bargellini et al	2014	Italy	22	Sorafenib	4 (18.2%)	68.3±8.2 §	NR	A: 22	B: 12 C: 10	>3: 11	50±37§
Ronot et al	2014	France	64	Sorafenib	8 (12.5%)	62 (37-77) ‡	NR	A: 51 B: 13	B: 20 C: 44	NR	NR
Salvaggio et al	2014	USA	17	Sorafenib	5 (29.4%)	69 (58-79) ‡	0: 8 1: 9	A: 16 B: 1	A: 3 B: 4 C: 10	NR	NR
McNamara et al	2015	Canada	30 (21)	Axitinib	9 (30.0%)	64 (18-78) ‡	0: 9 1: 21	A: 22 B: 8	C: 30	NR	NR
Takada et al	2015	Japan	191 (175)	Sorafenib	78 (40.8%)	72 (34–88) ‡	0: 141 1: 47 2: 3	A: 179 B: 12	A: 11 B: 85 C: 95	NR	NR

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Gavanier et al	2016	France	60	Sorafenib	6 (10.0%)	67 (39–79) ‡	NR	A: 42 B: 13 C: 5	B: 12 C: 48	NR	NR
Ikeda et al	2017	Japan	46 (42)	Lenvatinib	33 (71.7%)	66.5 (37–80) ‡	0: 38 1: 8	A: 45 B: 1	B: 19 C: 27	NR	NR
Pelosof et al	2018	USA	379	Regorafenib	46 (12.1%)	64 (19-85) ‡	0: 251 1: 128	A: 373 B: 5 C: 1	A: 1 B: 53 C: 325	NR	NR
Kaneko et al	2020	Japan	40	Lenvatinib	4 (10.0%)	72 (52–87) ‡	0: 25 1: 15	A: 38 B: 2	B: 12 C: 28	NR	NR
Kawamura et al	2020	Japan	51	Lenvatinib	16 (31.4%)	74 (45–91) ‡	0: 48 1: 3	A:51	A: 5 B: 23 C: 23	NR	31.8 (11.0–112.7) ‡
Kuzuya et al	2020	Japan	10	Ramucirumab	5 (50.0%)	76 (42-89) ‡	0: 7 1: 3	A: 9 B: 1	B: 6 C: 4	<4: 2 >4: 8	<30: 8 >30: 2
Maruta et al	2020	Japan	152 (131)	Lenvatinib	24 (15.8%)	>73:74 (49%)	≤1: 142 >1: 10	A: 132 B: 20	C: 99	>7: 70	>50: 51
Yamamichi et al	2020	Japan	22	Sorafenib	2 (9.1%)	76 (50–86) ‡	0: 19 1: 3	A:22	C: 22	NR	NR

† N = Number of included patients; n = Number of patients evaluated by RECIST 1.1 and mRECSIT criteria.

‡ Data are medians, with interquartile ranges in parentheses.

§ Data are means ± standard deviations.

ECOG PS, Eastern Cooperative Oncology Group performance status; BCLC stage, Barcelona Clinic Liver Cancer stage.

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Table 2. Time interval and imaging examinations

Included Trials	Exam	Time interval	Response considered
Spira et al	MRI	At baseline and follow-up every 8 weeks (range, 2–19 weeks; mean, 7.6 weeks).	Target response
Edeline et al	CT	1 CT scan in the first and second months and every 2 months thereafter.	Target response
Moschouris et al	CT/MRI	1-5 days prior to the initiation of antiangiogenetic treatment; follow-up studies were performed approximately every 2 months (range: 7-10 weeks) after the first dose of the drug.	Target response
Kawaoka et al	CT/MRI	At 8 weeks from the date of administration of sorafenib.	Overall response
Arizumi et al	CT/MRI	Every 4–6 weeks during and after treatment	Overall response
Bargellini et al	CT	4 ± 2 weeks before and 8 ± 2 weeks after initiation of sorafenib treatment	Overall response
Ronot et al	CT	6 weeks before sorafenib and had the first tumor evaluation with a second CT scan within 1–3 months after sorafenib initiation.	Overall response
Salvaggio et al	CT/MRI	Baseline examinations were performed at a median of 30 days (range 28–36 days) before the start of treatment. Follow-up imaging study available was performed after a median of 103 days (range 55–617 days).	Target response
McNamara et al	CT	Tumor response was assessed every 8 weeks via CT.	Overall response
Takada et al	CT	Within 1 month of commencing treatment and every 1–2 months during treatment.	Overall response
Gavanier et al	CT	Within 6 weeks before sorafenib administration; and Imaging available during sorafenib therapy (>4 weeks after initiation).	Overall response
Ikeda et al	CT/MRI	Tumor response was evaluated every 8 weeks	Overall response
Pelosof et al	CT/MRI	Tumor assessments were performed every 6 weeks for the first eight cycles, then every 12 weeks thereafter.	Overall response
Kaneko et al	CT	CT was performed at baseline and every 4–8 weeks after LEN administration.	Overall response
Kawamura et al	CT	We assessed the best tumor response during 2–12 weeks.	Overall response
Kuzuya et al	CT	CT examination was performed with a predetermined schedule at baseline and at 6 weeks after ramucirumab initiation.	Overall response

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Maruta et al	CT/MRI	Every 1–2 months after starting treatment for the evaluation of tumor response.	Overall response
Yamamichi et al	CT	CT was performed at baseline (before initiation of treatment) and at every 2–3 months afterward.	Overall response
Murakami et al	CT	CT was performed at baseline (before initiation of treatment) and at every 2–3 months afterward.	Overall response

CT, computed tomography; MRI, magnetic resonance imaging.

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Table 3. Inter-method agreement between RECIST 1.1 and mRECIST criteria.

Tumor response by RECIST1.1	Tumor response by mRECIST				Total
	CR	PR	SD	PD	
CR	6	0	0	0	6
PR	2	18	0	0	20
SD	4	38	166	5	213
PD	1	4	6	111	122
Total	13	60	172	116	361

RECIST 1.1, Response Evaluation Criteria in Solid Tumors version 1.1; mRECIST, modified Response Evaluation Criteria in Solid Tumors; CR, complete response; PR, partial response; SD, stable disease; PD, progressive disease.

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Table 4. Response assessment according to RECIST 1.1 and mRECIST criteria.

Study	N	Criterion	CR	PR	SD	PD
Spira et al	25	RECIST1.1	1	0	18	6
		mRECIST	1	11	9	4
Edeline et al	53	RECIST1.1	0	1	42	10
		mRECIST	2	10	30	11
Moschouris et al	21	RECIST1.1	0	1	16	4
		mRECIST	2	6	11	2
Kawaoka et al	49	RECIST1.1	1	1	30	17
		mRECIST	2	4	26	17
Arizumi et al	156	RECIST1.1	3	12	71	70
		mRECIST	6	30	55	65
Bargellini et al	22	RECIST1.1	0	1	5	16
		mRECIST	0	4	5	13
Ronot et al	64	RECIST1.1	2		43	19
		mRECIST	18		29	17
Salvaggio et al	17	RECIST1.1	0	2	10	5
		mRECIST	0	3	10	4
McNamara et al	21	RECIST1.1	0	2	19	0
		mRECIST	1	6	14	0
Takada et al	175	RECIST1.1	4	11	80	80
		mRECIST	5	20	72	78
Gavanier et al	60	RECIST1.1	0	2	28	30
		mRECIST	0	4	27	29
Ikeda et al.	42	RECIST1.1	0	11	25	6
		mRECIST	0	17	19	6
Pelosof et al.	379	RECIST1.1	0	25	354	
		mRECIST	2	38	339	
Kaneko et al.	40	RECIST1.1	1	9	21	9
		mRECIST	3	12	9	4
Kawamura et al	51	RECIST1.1	0	26	21	4
		mRECIST	6	32	9	4
Kuzuya et al.	10	RECIST1.1	0	0	8	2
		mRECIST	0	1	7	2
Maruta et al.	131	RECIST1.1	2	22	78	29
		mRECIST	3	59	42	27
Yamamichi et al	22	RECIST1.1	1	1	12	8
		mRECIST	1	1	7	13
Murakami et al.	27	RECIST1.1	0	0	16	11
		mRECIST	1	2	13	11

mRECIST, modified Response Evaluation Criteria in Solid Tumors; RECIST 1.1, Response Evaluation

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Criteria in Solid Tumors version 1.1; CR, complete response; PR, partial response; SD, stable disease; PD, progressive disease.

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Figure 1. Flow diagram of study selection

Figure 2. Forest plot for HR for overall survival (responders vs non-responders) according to mRECIST

Figure 3. Forest plot for HR for overall survival (responders vs non-responders) according to RECIST1.1

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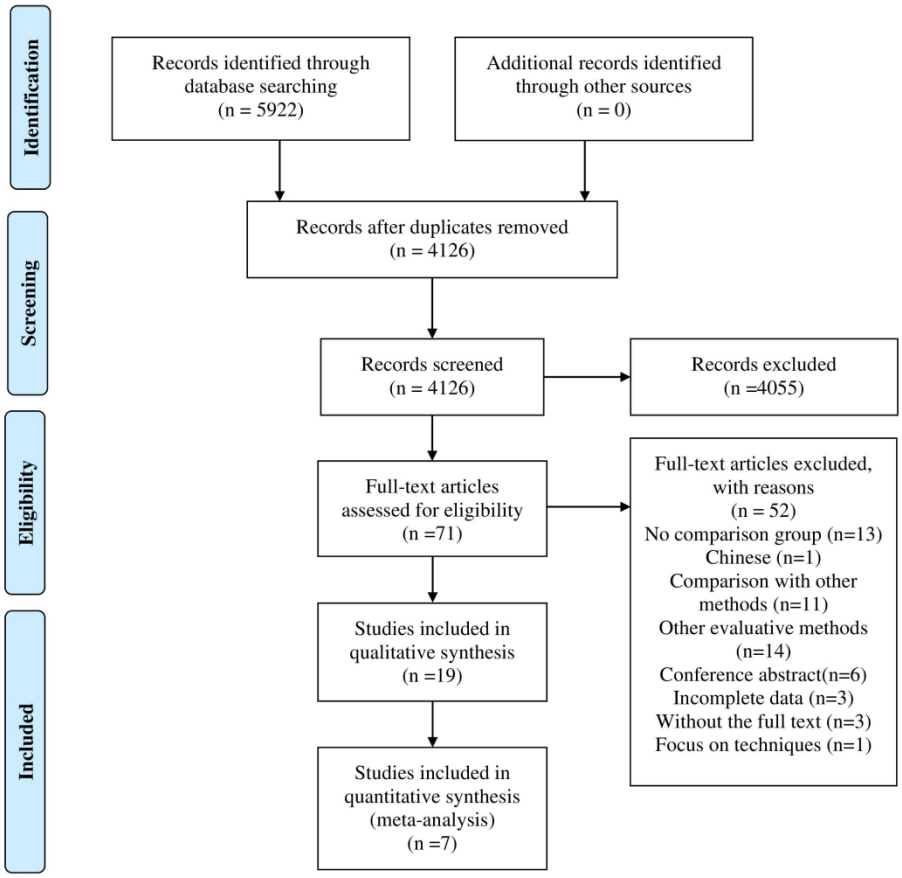

Identification
Screening
Eligibility
Included
Records identified through database searching (n = 5922)
Additional records identified through other sources (n = 0)
Records after duplicates removed (n = 4126)
Records screened (n = 4126)
Records excluded (n =4055)
Full-text articles assessed for eligibility (n =71)
Full-text articles excluded, with reasons (n = 52)
No comparison group (n=13)
Chinese (n=1)
Comparison with other methods (n=11)
Other evaluative methods (n=14)
Conference abstract(n=6)
Incomplete data (n=3)
Without the full text (n=3)
Focus on techniques (n=1)
Studies included in qualitative synthesis (n =19)
Studies included in quantitative synthesis (meta-analysis) (n =7)

Figure 1

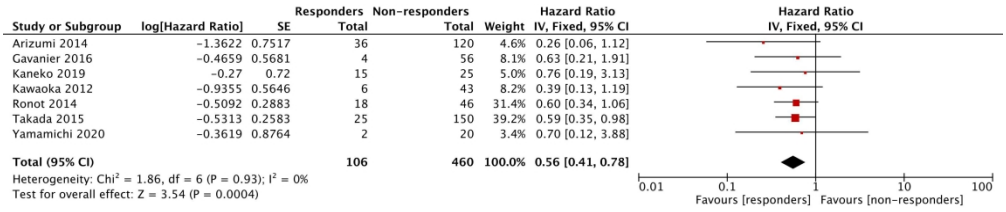

Responders
Non-responders
Hazard Ratio
Hazard Ratio
Study or Subgroup
log[Hazard Ratio]
SE
Total
Total
Weight
IV, Fixed, 95% CI
IV, Fixed, 95% CI
Arizumi 2014 -1.3622 0.7517 36 120 4.6% 0.26 [0.06, 1.12]
Gavanier 2016 -0.4659 0.5681 4 56 8.1% 0.63 [0.21, 1.91]
Kaneko 2019 -0.27 0.72 15 25 5.0% 0.76 [0.19, 3.13]
Kawaoka 2012 -0.9355 0.5646 6 43 8.2% 0.39 [0.13, 1.19]
Ronot 2014 -0.5092 0.2883 18 46 31.4% 0.60 [0.34, 1.06]
Takada 2015 -0.5313 0.2583 25 150 39.2% 0.59 [0.35, 0.98]
Yamamichi 2020 -0.3619 0.8764 2 20 3.4% 0.70 [0.12, 3.88]
Total (95% CI) 106 460 100.0% 0.56 [0.41, 0.78]
Heterogeneity: Chi² = 1.86, df = 6 (P = 0.93); I² = 0%
Test for overall effect: Z = 3.54 (P = 0.0004)
0.01
0.1
1
10
100
Favours [responders]
Favours [non-responders]

Figure 2

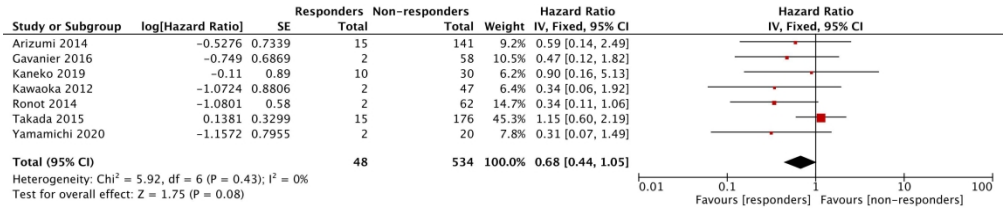
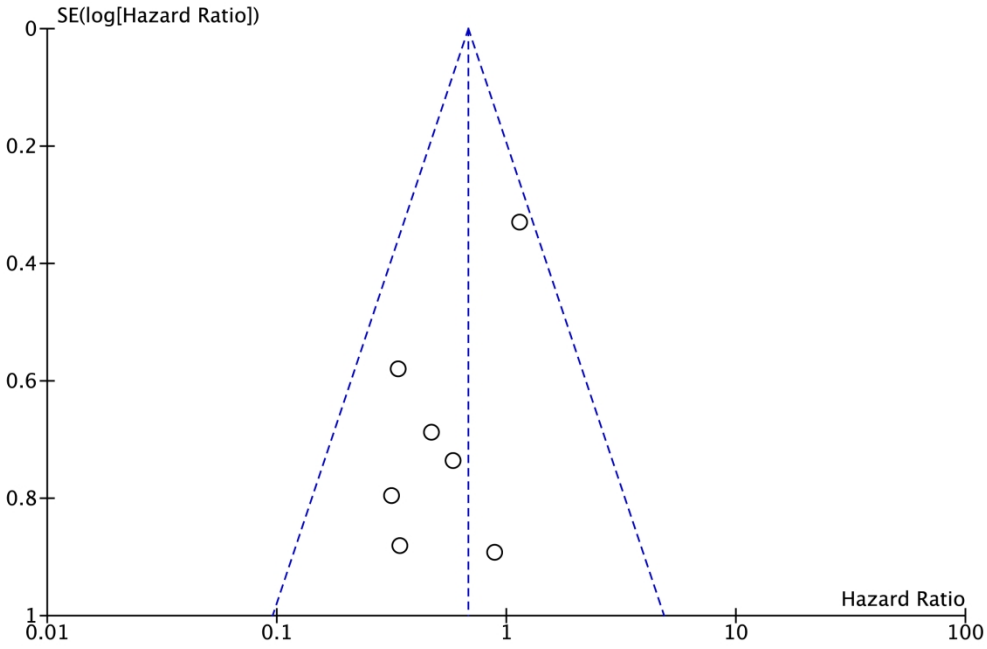
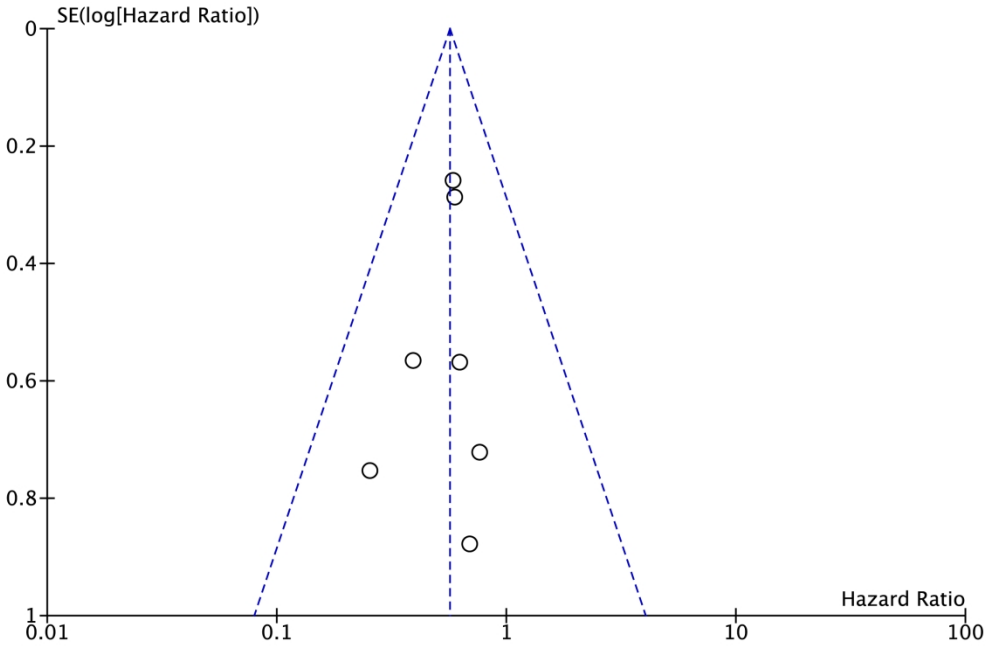

Responders Non-responders Hazard Ratio Hazard Ratio
Study or Subgroup log[Hazard Ratio] SE Total Total Weight IV, Fixed, 95% CI IV, Fixed, 95% CI
Arizumi 2014 -0.5276 0.7339 15 141 9.2% 0.59 [0.14, 2.49]
Gavanier 2016 -0.749 0.6869 2 58 10.5% 0.47 [0.12, 1.82]
Kaneko 2019 -0.11 0.89 10 30 6.2% 0.90 [0.16, 5.13]
Kawaoka 2012 -1.0724 0.8806 2 47 6.4% 0.34 [0.06, 1.92]
Ronot 2014 -1.0801 0.58 2 62 14.7% 0.34 [0.11, 1.06]
Takada 2015 0.1381 0.3299 15 176 45.3% 1.15 [0.60, 2.19]
Yamamichi 2020 -1.1572 0.7955 2 20 7.8% 0.31 [0.07, 1.49]
Total (95% CI) 48 534 100.0% 0.68 [0.44, 1.05]
Heterogeneity: Chi² = 5.92, df = 6 (P = 0.43); I² = 0%
Test for overall effect: Z = 1.75 (P = 0.08)
0.01 0.1 1 10 100
Favours [responders] Favours [non-responders]

Figure 3



211x141mm (300 x 300 DPI)



211x141mm (300 x 300 DPI)



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	#1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	#2; #3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	#5; #6
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	#6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	#7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	#7
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	#7
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	#7
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	#8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	#8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	#8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	#8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	#9
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	#9
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	#9
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	#9

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	#9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	#9
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	#9
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	#9
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	#10
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	#10
Study characteristics	17	Cite each included study and present its characteristics.	#11
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	#11
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	#12
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	#12; #13
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	#12; #13
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	#13
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	#13
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	#13
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	#14; #15
	23b	Discuss any limitations of the evidence included in the review.	#16
	23c	Discuss any limitations of the review processes used.	#16

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	23d	Discuss implications of the results for practice, policy, and future research.	#16
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	#3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	#3
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	#1
Competing interests	26	Declare any competing interests of review authors.	#1
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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RECIST 1.1 versus mRECIST for assessment of tumour response to molecular targeted therapies and disease outcomes in patients with hepatocellular carcinoma: a systematic review and meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-052294.R1
Article Type:	Original research
Date Submitted by the Author:	15-Feb-2022
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Primary Subject Heading:	Oncology
Secondary Subject Heading:	Gastroenterology and hepatology
Keywords:	Hepatobiliary disease < GASTROENTEROLOGY, Hepatology < INTERNAL MEDICINE, Hepatobiliary tumours < ONCOLOGY, Magnetic resonance imaging < RADIOLOGY & IMAGING, Computed tomography < RADIOLOGY & IMAGING

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RECIST 1.1 versus mRECIST for assessment of tumour response to molecular targeted therapies and disease outcomes in patients with hepatocellular carcinoma: a systematic review and meta-analysis

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Competing interests:

None of the authors have identified a conflict of interest.

Funding:

This work was funded by the National Natural Science Foundation of China (No.

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81770607 and 81772626)

Contributorship statement:

Yuping Bai and Hongli Yu contributed to to the design of the study, data collection, data analysis and article writing; Xiaoyu Xie contributed to data collection; Yuemin Feng and Yao Yang contributed equally to the data analysis. Qiang Zhu contributed to the conception and design of the study and critical revision. All authors contributed to the revision and gave final approval of the article.

Data availability statement:

All data included in this study are available upon request by contacting with the corresponding author.

RECIST 1.1 versus mRECIST for assessment of tumour response to molecular targeted therapies and disease outcomes in patients with hepatocellular carcinoma: a systematic review and meta-analysis

Abstract

Objectives Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST 1.1) and modified RECIST (mRECIST) are commonly used to assess tumor response. Which one is better to evaluate efficacy after molecular targeted therapies in hepatocellular carcinoma (HCC) patients is still controversial. A systemic review was

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performed to compare the objective response rate (ORR) and disease control rate (DCR) and a meta-analysis was conducted to compare the correlation between objective response and overall survival (OS).

Design Systematic review and meta-analysis using the Grading of Recommendations Assessment, Development, and Evaluation approach.

Data sources EMBASE, PubMed, Web of Science and Cochrane Library were searched through December 31, 2021.

Eligibility criteria We included studies assessing the efficacy of molecular targeted therapy for HCC according to both RECIST 1.1 and mRECIST.

Data extraction and synthesis Two investigators extracted data independently. The consistency between RECIST 1.1 versus mRECIST is measured by the k coefficient. Hazard ratio (HRs) with corresponding 95% confidence intervals (CIs) were used for meta-analysis.

Results 23 studies comprising 2,563 patients were included in systematic review. The ORR according to mRECIST is higher than RECIST1.1 (15.9% versus 7.8%, p<0.001). The DCR is similar (68.4% versus 67.2%, p=0.5). The agreement of tumor response is moderate for objective response (k=0.499) and perfect for progressive disease (k=0.901), calculated from 8 studies including 372 patients. OS was significantly longer in response group than non-response group according to mRECIST (HR=0.56, 95% CI=0.41-0.78, p=0.0004) calculated from 7 studies including 566 patients, however the RECIST1.1 couldn't distinguish the OS well (HR=0.68, 95%CI, 0.44-1.05, p=0.08). Subgroup analusis by type of treatment was conducted.

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Conclusions mRECIST may be more accurate than RECIST1.1 in assessing objective response rate after molecular targeted therapies in HCC patients and can better assess the prognosis. However, the performance of both criteria in assessing disease progression is identical.

PROSPERO registration number CRD42020200895.

Ethics approval Ethics approval is not required in this meta-analysis.

Key Words:

Hepatocellular carcinoma; Molecular targeted therapies; Modified Response Evaluation Criteria in Solid Tumors; Response Evaluation Criteria in Solid Tumors version 1.1; Meta-analysis

Strengths and limitations of this study

1. Quantitative analysis of RECIST1.1 and mRECIST to assess the relationship between tumour response and overall survival after molecular targeted therapies in patients with HCC.
2. Reliable methodological and statistical procedures were applied.
3. This study is limited by a small number of papers after screening according to inclusion and exclusion criteria.

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4. The variable intervals between follow-up imaging results could be a source of heterogeneity.

Abbreviations:

BCLC, Barcelona Clinic Liver Cancer; CI, confidence interval; CR, complete response; CT, computed tomography; DCR, disease control rate; EASL, European Association for the Study of the Liver; ECOG PS, Eastern Cooperative Oncology Group performance status; ESMO, European Society for Medical Oncology; HCC, hepatocellular carcinoma; HR, hazard ratio; mRECIST, modified Response Evaluation Criteria in Solid Tumors; MRI, magnetic resonance imaging; NOS, Newcastle-Ottawa scale; OR, objective response; ORR, objective response rate; OS, overall survival; PD, progressive disease; PET, positron emission tomography; PR, partial response; PRISMA, Preferred Reporting Items for Systematic reviews and Meta-Analyses; RECIST 1.0, Response Evaluation Criteria in Solid Tumors version 1.0; RECIST 1.1, Response Evaluation Criteria in Solid Tumors version 1.1; SD, stable disease; WHO, World Health Organization.

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Introduction

Hepatocellular carcinoma (HCC) is the sixth most common cancer and the fourth leading cause of cancer-related mortality worldwide [1]. There have been significant advances in treatment for HCC over the past decade. Available treatment options include surgical resection, liver transplantation, ablative techniques, transarterial chemoembolization, transarterial radioembolization, radiotherapy and molecular targeted therapies [2]. Molecular targeted therapies are indicated for patients with advanced tumors or earlier stage unsuitable for surgical resection or locoregional therapies [3]. It has already been confirmed that molecular targeted therapies can improve survival in patients with HCC due to their unique antiproliferative and antiangiogenic function [4]. The accurate assessment of therapeutic efficacy of molecular targeted therapies is essential for routine anti-cancer treatment as well as clinical trials.

Radiological evaluation of tumor response is a well-recognized surrogate endpoint in the assessment of therapeutic efficacy of molecular targeted therapies in patients with HCC [5], which is crucial to help identify potentially resistant patients, avoiding unnecessary toxicities. Tumor response was initially measured according to the World Health Organization (WHO) criteria and Response Evaluation Criteria in Solid Tumors version 1.0 (RECIST 1.0) guideline [6 7]. Nevertheless, they have been proven to correlate poorly with survival outcomes in HCC patients and provide insufficient guidance on treatment options [8-10]. Nowadays, the RECIST 1.1 and modified RECIST (mRECIST) criteria are the most commonly used criteria to assess tumor response. The major

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changes in RECIST 1.1 include the reduction in the number of target lesions and the augmented definition of disease progression [11], which relies on the change in the sum of the greatest diameters. The mRECIST has been developed which differ from RECIST 1.1 in that the target lesion measured is not the whole lesion but only the viable tumor, defined as the contrast-enhanced portion of the tumor on hepatic arterial phase images[8].

European Association for the Study of the Liver (EASL) and European Society for Medical Oncology guidelines (ESMO) suggested applying mRECIST or RECIST 1.1 in patients with HCC treated with molecular targeted therapies [3] [12]. However, National Comprehensive Cancer Network guideline indicated that validated criterion to evaluate tumor response to molecular targeted therapies between the two criteria is needed [13]. Besides, several studies demonstrated that overall survival can be predicted more accurately by mRECIST than RECIST 1.1, since the latter is not capable of assessing therapy induced intratumoral necrosis [14] [15]. On the contrary, another study observed both methods provided correlation with OS equally[16]. Which set of criteria is better to assess response to molecular targeted therapies remains controversial.

We perform this systematic review to compare the efficacy of RECIST1.1 and mRECIST in assessing tumor response after molecular targeted therapies in patients with HCC and to quantitatively determine which criterion correlates better with prognosis.

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Material and methods

Protocol and guidance

This study was reported according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) and registered in PROSPERO (CRD42020200895).

Patient and public involvement

Patients and the public were not involved in this meta-analysis.

Search strategy

A comprehensive search of PubMed, EMBASE, Web of Science, and the Cochrane Library from inception through December 31, 2021 was performed. The following Mesh terms and text words were confined to the title or abstract: “RECIST”, “mRECIST”, “Response Evaluation Criteria in Solid Tumors”, “liver cancer” and “hepatocellular carcinoma”. The detailed search strategy is included in the supplementary materials (Table S1). The reference lists of relevant articles were also searched for other eligible studies.

Selection of studies

Two reviewers (Bai and Yu) independently assessed articles for eligibility, and discrepancies were resolved by a consensus and confirmed by another author Yang. To

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be eligible for inclusion, studies had to meet the following criteria: (1) the diagnosis of HCC was based on pathology or radiological findings, in accordance with the criteria of practice guidelines; (2) patients with HCC must be treated with molecular targeted therapies; (3) response assessment after molecular targeted therapies was evaluated according to both RECIST 1.1 and mRECIST criteria; (4) available data about overall survival and k coefficient or sufficient information to calculate it. General exclusion criteria were: (1) presence of an additional primary malignancy in other organ; (2) patients with HCC received other therapies; (3) case analysis, letters, reviews and expert opinions; (4) studies with incomplete data; (5) published in languages other than English with no translation.

Quality assessment

The Newcastle-Ottawa scale (NOS) was used to assess the quality of the studies, this scale consists of three factors: the selection of patients, comparability of the study groups, and assessment of outcome[17]. The maximum total score on this scale is 9 and studies with scores ≥6 were defined as high-quality studies.

Data extraction

Two investigators (Yang and Yu) assessed and extracted data from all eligible studies independently, and discrepancies were resolved by a consensus and confirmed by another author Bai. By reading the full texts of the selected studies, two investigators extracted the following data: name of all authors, year of publication, number of

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enrolled patients, age, sex, ECOG PS, Child-Pugh score, BCLC stage, tumor number, tumor size, type of treatment, reported hazard ratio (HR) for overall survival (OS) according to mRECIST and RECIST 1.1 criteria and k coefficient of concordance in each study.

Tumor response assessment

Evaluation of tumor response according to the RECIST 1.1 was defined as follows: complete response (CR) is the disappearance of all target lesions; partial response (PR) is at least a 30% decrease in the sum of the diameters of the target lesions; progressive disease (PD) is at least a 20% increase in the sum of the longest diameters of target lesions or the appearance of one or more new lesions; stable disease (SD) is neither sufficient shrinkage to qualify for PR nor sufficient increase to qualify for PD. According to mRECIST, CR is defined as the disappearance of any intratumoral arterial enhancement in all target lesions; PR is at least a 30% decrease in the sum of diameters of viable target lesions, taking as reference the baseline sum of the diameters of target lesions; PD is an increase of at least 20% in the sum of the diameters of viable target lesions, taking as reference the smallest sum of the diameters of viable target lesions recorded since the treatment started; SD is any cases that do not qualify for either PR or PD. Objective response (OR) included both CR and PR, and disease control included CR, PR and SD[8].

Statistical analysis

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Inter-method agreement between similar categorical items of the two criteria was measured using the k coefficient. The agreement was interpreted as poor ($k < 0$), slight ($k = 0$ to 0.20), fair ($k = 0.21$ to 0.40), moderate ($k = 0.41$ to 0.60), substantial ($k = 0.61$ to 0.80), and almost perfect ($k > 0.80$)[18]. The objective response rate (ORR) and disease control rate (DCR) between the two criteria were compared by the chi-square tests with the significance at $p<0.05$. Hazard ratio (HRs) with corresponding 95% confidence intervals (CIs) were performed to estimate the relationship between the ORR and OS of patients with HCC. The HRs were extracted from the text or from the K-M curves by the software Engauge Digitizer. The heterogeneity was quantified using the I^2 statistic. A fixed-effects model was used to analyze the results if the $I^2 \leq 50\%$, whereas the random effects model is applied if the $I^2>50\%$ among the included studies, funnel plots and Egger's test were used to grossly exclude publication bias. All extracted data analyses were performed with Review Manager 5.4.1 and STATA 15.1 and SPSS 24 software.

Results

Eligible studies for analysis

A total of 7281 studies were retrieved from the electronic database search. After removing duplicates, 5046 titles and abstracts were further examined. 114 publications underwent full text review to determine their eligibility for the meta-analysis and 91 were excluded. 49 studies were excluded because they applied only the mRECIST or RECIST1.1 criteria alone. 14 studies were excluded because using other imaging

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examinations or other evaluative methods, such as positron emission tomography (PET) imaging, contrast-enhanced ultrasonography and RECIST1.0. 12 studies compared with other methods, such as Choi and EASL criteria et al. Six studies were excluded because they were all conference abstracts. Three studies were excluded for unavailable for full text. Five studies were excluded for lacking of original data and had insufficient data for extraction. One study focused on manual and automatically extracted measurements. One research was written in Japanese.

Finally, 23 studies including 2563 patients comparing tumor response between the RECIST 1.1 and mRECIST criteria were included [14-16 19-38] (**Figure 1**).

Summary of baseline characteristics

2563 patients form 23 studies were included in qualitative analysis (**Table 1**). 1325 patients treated with sorafenib [14-16 19-25 27 34 36 37], 21 patients treated with axitinib [26], 828 patients treated with lenvatinib [28 30 31 33 35 36 38], 379 patients treated with regorafenib [29] and 10 patients treated with ramucirumab [32]. Most studies included Child-Pugh class A and a minority of Child-Pugh class B patients, only two studies included a small percentage of Child-Pugh class C patients [27 29].

Due to lacking of survival data, seven studies including 566 patients were finally included in this meta-analysis. Six of the studies[14 15 21 23 27 34] included 526 patients treated with sorafenib and one study[30] included 40 patients treated with lenvatinib.

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Evaluation of tumor response was performed according to the RECIST 1.1 and mRECIST criteria and assessment of response was carried out by contrast-enhanced spiral computed tomography (CT) or gadolinium-enhanced magnetic resonance imaging (MRI) after 4–8 weeks from treatment, depending on each study (**Table 2**).

Risk of bias within studies

All of 23 studies had good quality. The quality of included studies assessed by NOS was 6 to 8.

Comparison of tumor response between the RECIST 1.1 and mRECIST criteria.

Table 3 shows the tumor response assessed by RECIST 1.1 and mRECIST after molecular targeted therapies in the 23 considered studies. The ORR according to mRECIST was significantly higher than RECIST1.1 (15.9% versus 7.8%, $p<0.001$). For disease control rate (DCR), four study was considered not eligible for its incomplete data [29 36-38]. The DCR was similar according to mRECIST and RECIST1.1 (68.4% versus 67.2%, $p=0.5$).

The agreement and disagreement of tumor response of the two criteria were described in **Table 4**, which could be available or calculated from 8 studies including 372 patients [15 16 20-22 25 26 38]. For objective response, the agreement of tumor response between the two criteria was moderate (k=0.499). Of 218 patients with SD according to RECIST1.1,

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45 patients were reclassified to objective response according to mRECIST. For disease control, the agreement of tumor response between the two criteria was almost perfect (k=0.901). Of 116 patients with PD according to mRECIST, only 5 patients were reclassified to SD according to RECIST1.1.

Subgroup analysis was performed based on therapeutic agents. Of the total of eight studies included in the consistency test, six studies[15 16 20-22 25 26] including 340 patients receiving sorafenib, one study[26] with 11 patients treated with Lenvatinib and one study[38] with 21 patients treated with axitinib. Limited by the sample size, we only performed an analysis of concordance in the sorafenib group. For objective response, the agreement of tumor response between the two criteria was moderate (k=0.446). For disease control, the agreement of tumor response between the two criteria was almost perfect (k=0.897).

Survival analysis according to the RECIST 1.1 and mRECIST criteria.

Of the nineteen articles, 12 studies were excluded due to lack of survival data. Finally, seven studies including 566 patients were included in this meta-analysis. [14 15 21 23 27 30 34]. The ORR according to RECIST 1.1 and mRECIST criteria was 7.79% and 15.93%, respectively. According to mRECIST, OS was significantly longer in patients with response than patients with non-response (HR=0.56, 95% CI=0.41-0.78, p=0.0004) (**Figure 2**), with no significant heterogeneity among the studies (I^2=0, p=0.93). In contrast, RECIST1.1 could not distinguish well between the responders and the non-

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responders for OS (HR=0.68, 95%CI, 0.44-1.05, p=0.08) (**Figure 3**), with no significant heterogeneity among the studies ((I^2=0, p=0.43). Funnel plots for both RECIST 1.1 and mRECIST did not show asymmetry (**Supplementary figure 1-2**). Egger's test also showed no clear evidence of publication bias (p=0.052 for RECIST1.1 and p=0.503 for mRECIST).

Subgroup analysis was performed based on therapeutic agents. Of the total of seven studies included in the survival analysis, six studies[14 15 21 23 27 34] including 526 patients receiving sorafenib and one study[30] with 40 patients treated with lenvatinib.

As shown in Figure 4 and Figure 5, among patients receiving sorafenib, when mRECIST was used as an evaluation criterion, OS was significantly longer in patients who responded (HR=0.56, 95%CI=0.40, 0.77), while using RECIST1.1 as the evaluation criterion failed to clearly distinguish between responder and non-responder (HR=0.67, 95% CI=0.42, 1.04). Possibly limited by sample size, tumour response assessed using mRECIST or RECIST1.1 did not differentiate well between OS in responders and non-responders for patients receiving lenvatinib (HR=0.76, 95%CI 0.19, 3.13 v.s. HR=0.90, 95%CI=0.16, 5.13). No significant heterogeneity was found in the subgroup analysis.

Discussion

Tumor response assessment is criticalin the management of cancer.It serves as a guide

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to clinical practice and asa surrogate endpoint for evaluating efficacy in clinical studies [39 40]. In particular, an increasing number of patients with HCC have been treated with molecularly targeted therapies in recent years.

Since new molecular targeted agents exert antitumoral activity by inducing tumor necrosis, with rare changes in volume shrinkage, traditional WHO and RECIST criteria do not always represent an appropriate tool for response evaluation. Anatomic imaging alone may have limitations, particularly in assessing the activity of targeted therapies which stabilize diseases. This promoted the development of the mRECIST for a response that incorporated treatment-induced tumor necrosis by dynamic imaging. In this study, we compared the effectiveness of the RECIST 1.1 and mRECIST criteria in assessing the efficacy of molecular targeted therapies in patient with HCC.

We investigated the concordance between the RECIST 1.1 and mRECIST criteria for the assessment of tumor response in patients with HCC treated with molecular targeted therapies. Our results showed that there was a considerable discrepancy in the assessment of objective responses between the RECIST 1.1 and mRECIST criteria. When adopting the mRECIST, the ORR was significantly higher, suggesting that mRECIST better identifies the response of hepatocellular carcinoma after molecularly targeted therapy. There are several possible reasons. Firstly, molecular targeted therapies are based on the inhibition of several pro-angiogenic signaling pathways, which stimulating angiogenesis, are responsible of the characteristic hyper-vascular

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pattern of HCC lesions[41]. The therapeutic response after molecularly targeted therapy is closely associated with structural changes, mainly including decreased vascularization and increased tissue necrosis or cavitation, but it is not always reflected in the reduction in tumor size[42]. Secondly, HCC and cirrhosis coexist in more than 80% of cases. The inherent pathogenic factors and haemodynamic changes of cirrhosis may mimic or mask intrahepatic tumours[43].

From a clinical perspective, clinicians need to accurately distinguish between progressive disease and disease control, and thus make clinical decisions to switch from first-line to second-line treatment when disease progresses. We also found that there was an excellent agreement in the assessment of the disease progression between the RECIST 1.1 and mRECIST criteria, both of the criteria are equally able to discriminate progressors and non-progressors and thus equally able to give appropriate guidance for clinical decision making, which is the most relevant parameter in clinical practice. And we presume that this consistency is due to the fact that disease progression appears to involve an increase in vascularisation, which transforms into an increase in lesion dimension. In general, it is assumed that inter-operator variability can affect the interpretation of the same image, even when guided by the same evaluation criteria. In particular, evaluation based on mRECIST addresses the subjectivity of the reviewer. However, it has been shown that in the evaluation of disease control, there is still a high level of agreement between the results obtained by experts and those without specialist training in liver imaging ($k=0.737 \pm 0.114$)[44].

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Our results also show that mRECIST can be of help in predicting overall survival in patients receiving molecular targeted therapies. Those patients with objective response having significantly better survival outcome compared to patients who only achieve SD or PD. However, the OS of those classified as objective response by RECIST1.1 is not significantly different from that of non-responders. Edeline et al demonstrated that in the 79.2% of patients classified as stable by RECIST 1.1, the use of mRECIST enabled the prediction of different prognostic subgroups with a significantly better median OS of 17.1 months for responding patients compared with 9.7 months for stable patients and 3.7 months in patients who had a PD[16]. Our results suggest that mRECIST may offer a suitable alternative to RECIST in Phase II clinical trials, in which detection of an efficacy signal is paramount. However, as previously mentioned, mRECIST did not demonstrate superiority in guiding the replacement of second-line therapeutic agents.

In addition, We find that most of included patients were classified as PD. Patients with PD always have a poor outcome. Treatment beyond radiological progression is not warranted and that patients should be actively monitored for radiological progression rather than waiting for symptomatic progression. A recent review indicated that 'Progressive disease' concept includes different patterns of progression leading to different prognosis. The reason for the imperfect correlation between surrogate end point and OS likely relies on the basis that not all patterns of progressions are equal in terms of prognostic implications. In a brilliant paper, Reig et al. demonstrated that the

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appearance of new extrahepatic lesions has a far worse prognostic impact than the enlargement of pre-existing lesions or the appearance of new intrahepatic nodules [45]. Thus, a careful evaluation of the progression pattern is indeed required in clinical practice before switching to a second line treatment.

There are several limitations in our study. First, most of the included studies were retrospective. Second, this study included heterogeneous patients with different kinds of therapeutic agents and the variable interval between follow-up imaging examinations. It is necessary to verify these results in studies with larger homogeneous patients' cohort.

In conclusion, RECIST1.1 has similar efficacy to mRECIST in assessing disease progression with molecularly targeted drugs, but mRECIST is better at identifying objective responses. And mRECIST appears more appropriate than RECIST 1.1 to identify responders with long survival benefiting from molecular targeted therapies in patients with HCC.

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Table1. Baseline characteristics of include studies

Study	Year	Country	Patients, N (n) †	Treatment	Women, No. (%)	Age, y	ECOG PS	Child-Pugh class	BCLC stage	Tumor number	Tumor size (mm)
Spira et al[20]	2011	Germany	25	Sorafenib	2 (8.0%)	65 (42–75) ‡	0: 6 1 or 2: 19	A: 23 B: 2	NR	NR	NR
Murakami et al[19]	2011	Japan	27	Sorafenib	1 (3.7%)	63	NR	A: 23 B: 4	NR	NR	NR
Edeline et al[16]	2012	France	53	Sorafenib	5 (9.4%)	NR	0: 29 1: 24	A: 41 B: 12	B: 22 C: 31	NR	NR
Moschouris et al[22]	2012	UK	21	Sorafenib	2 (9.5%)	66.8±8.5§	NR	A: 10 B: 11	B: 9 C: 12	Solitary: 9 Multifocal: 12	NR
Kawaoka et al[21]	2012	Japan	66 (49)	Sorafenib	8 (12.1%)	63 (35-80) ‡	NR	A: 58 B: 8	0: 2 A: 1 B: 14 C: 49	1–2: 16 />3: 50	50 (8.3–194) ‡
Arizumi et al[23]	2014	Japan	156	Sorafenib	36 (23.1%)	73 (66-78) ‡	0: 150 1: 5 2: 1	A: 129 B: 27	A: 39 B: 36 C: 81	NR	NR
Bargellini et al[24]	2014	Italy	22	Sorafenib	4 (18.2%)	68.3±8.2§	NR	A: 22	B: 12 C: 10	>3: 11	50±37§
Ronot et al[14]	2014	France	64	Sorafenib	8 (12.5%)	62 (37-77) ‡	NR	A: 51 B: 13	B: 20 C: 44	NR	NR
Salvaggio et al[25]	2014	USA	17	Sorafenib	5 (29.4%)	69 (58-79) ‡	0: 8 1: 9	A: 16 B: 1	A: 3 B: 4 C: 10	NR	NR
McNamara et al[26]	2015	Canada	30 (21)	Axitinib	9 (30.0%)	64 (18-78) ‡	0: 9 1: 21	A: 22 B: 8	C: 30	NR	NR
Takada et al[15]	2015	Japan	191 (175)	Sorafenib	78 (40.8%)	72 (34–88) ‡	0: 141 1: 47 2: 3	A: 179 B: 12	A: 11 B: 85 C: 95	NR	NR
Gavanier et al[27]	2016	France	60	Sorafenib	6 (10.0%)	67 (39–79) ‡	NR	A: 42 B: 13 C: 5	B: 12 C: 48	NR	NR
Ikeda et al[28]	2017	Japan	46 (42)	Lenvatinib	33 (71.7%)	66.5 (37–80) ‡	0: 38 1: 8	A: 45 B: 1	B: 19 C: 27	NR	NR
Pelosof et al[29]	2018	USA	379	Regorafenib	46 (12.1%)	64 (19-85) ‡	0: 251 1: 128	A: 373 B: 5 C: 1	A: 1 B: 53 C: 325	NR	NR

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Kaneko et al[30]	2020	Japan	40	Lenvatinib	4 (10.0%)	72 (52–87) ‡	0: 25 1: 15	A: 38 B: 2	B: 12 C: 28	NR	NR
Kawamura et al[31]	2020	Japan	51	Lenvatinib	16 (31.4%)	74 (45–91) ‡	0: 48 1: 3	A:51	A: 5 B: 23 C: 23	NR	31.8 (11.0–112.7) ‡
Kuzuya et al[32]	2020	Japan	10	Ramucirumab	5 (50.0%)	76 (42-89) ‡	0: 7 1: 3	A: 9 B: 1	B: 6 C: 4	<4: 2 >4: 8	<30: 8 >30: 2
Maruta et al[33]	2020	Japan	152 (131)	Lenvatinib	24 (15.8%)	>73:74 (49%)	≤1: 142 >1: 10	A: 132 B: 20	C: 99	>7: 70	>50: 51
Yamamichi et al[34]	2020	Japan	22	Sorafenib	2 (9.1%)	76 (50–86) ‡	0: 19 1: 3	A:22	C: 22	NR	NR
He et al.[35]	2021	China	86	Lenvatinib	6(10.5)	>50: 44(51.2%)	0: 22 1: 64	A: 86	C: 86	1-3:9 >3:77	≤10:40 > 10:46
Nair et al.[36]	2021	USA	LEN 478/ SOR 476	Lenvatinib/ Sorafenib	LEN 73 (15.3) SOR 75 (15.7)	LEN<65: 56% SOR<65: 60%	LEN 0: 301 1: 177 SOR 0: 299 1: 177	LEN A: 475 B: 3 SOR A: 471 B: 5	LEN B: 104 C: 374 SOR B: 92 C: 384	LEN 1: 207 2: 167 3: 103 SOR 1: 207 2: 183 3: 86	NR
Salem et al.[37]	2021	USA	165(158)	Sorafenib	28(17.0)	64.4 ±10.9	NR	A: 165	NR	NR	NR
Yamashige et al.[38]	2021	Japan	11(22)	Lenvatinib	3(27.3)	67(59-83)	NR	A: 11	B: 6 C: 5	NR	NR

† N = Number of included patients; n = Number of patients evaluated by RECIST 1.1 and mRECSIT criteria.

‡ Data are medians, with interquartile ranges in parentheses.

§ Data are means ± standard deviations.

ECOG PS, Eastern Cooperative Oncology Group performance status; BCLC stage, Barcelona Clinic Liver Cancer stage.

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Table2. Time interval and imaging examinations

Included Trials	Exam	Time interval	Response considered
Spira et al[20]	MRI	At baseline and follow-up every 8 weeks (range, 2–19 weeks; mean, 7.6 weeks).	Target response
Edeline et al[16]	CT	1 CT scan in the first and second months and every 2 months thereafter.	Target response
Moschouris et al[22]	CT/MRI	1-5 days prior to the initiation of antiangiogenetic treatment; follow-up studies were performed approximately every 2 months (range: 7-10 weeks) after the first dose of the drug.	Target response
Kawaoka et al[21]	CT/MRI	At 8 weeks from the date of administration of sorafenib.	Overall response
Arizumi et al[23]	CT/MRI	Every 4–6 weeks during and after treatment	Overall response
Bargellini et al[24]	CT	4 ± 2 weeks before and 8 ± 2 weeks after initiation of sorafenib treatment	Overall response
Ronot et al[14]	CT	6 weeks before sorafenib and had the first tumor evaluation with a second CT scan within 1–3 months after sorafenib initiation.	Overall response
Salvaggio et al[25]	CT/MRI	Baseline examinations were performed at a median of 30 days (range 28–36 days) before the start of treatment. Follow-up imaging study available was performed after a median of 103 days (range 55–617 days).	Target response
McNamara et al[26]	CT	Tumor response was assessed every 8 weeks via CT.	Overall response
Takada et al[15]	CT	Within 1 month of commencing treatment and every 1–2 months during treatment.	Overall response
Gavanier et al[27]	CT	Within 6 weeks before sorafenib administration; and Imaging available during sorafenib therapy (>4 weeks after initiation).	Overall response
Ikeda et al[28]	CT/MRI	Tumor response was evaluated every 8 weeks	Overall response
Pelosof et al[29]	CT/MRI	Tumor assessments were performed every 6 weeks for the first eight cycles, then every 12 weeks thereafter.	Overall response
Kaneko et al[30]	CT	CT was performed at baseline and every 4–8 weeks after LEN administration.	Overall response
Kawamura et al[31]	CT	We assessed the best tumor response during 2–12 weeks.	Overall response
Kuzuya et al[32]	CT	CT examination was performed with a predetermined schedule at baseline and at 6 weeks after ramucirumab initiation.	Overall response

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Maruta et al[33]	CT/MRI	Every 1–2 months after starting treatment for the evaluation of tumor response.	Overall response
Yamamichi et al[34]	CT	CT was performed at baseline (before initiation of treatment) and at every 2–3 months afterward.	Overall response
Murakami et al[19]	CT	CT was performed at baseline (before initiation of treatment) and at every 2–3 months afterward.	Overall response
He et al[35]	CT/MRI	Upper abdomen-enhanced CT (or MRI) was performed at baseline and every 6 weeks (± 1 week).	Overall response
Nair et al[36]	CT/MRI	Enhanced CT or MRI was performed at baseline and every 8 weeks.	Overall response
Salem et al[37]	CT/MRI	Enhanced CT or MRI was performe at baseline and every 6 weeks.	Target response
Yamashige et al[38]	CT	Enhanced CT was performed at baseline and every 2-12 weeks.	Target response

CT, computed tomography; MRI, magnetic resonance imaging.

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Table3. Response assessment according to RECIST 1.1 and mRECIST criteria.

Study	N	Criterion	CR	PR	SD	PD
Spira et al[20]	25	RECIST1.1	1	0	18	6
		mRECIST	1	11	9	4
Edeline et al[16]	53	RECIST1.1	0	1	42	10
		mRECIST	2	10	30	11
Moschouris et al[22]	21	RECIST1.1	0	1	16	4
		mRECIST	2	6	11	2
Kawaoka et al[21]	49	RECIST1.1	1	1	30	17
		mRECIST	2	4	26	17
Arizumi et al[23]	156	RECIST1.1	3	12	71	70
		mRECIST	6	30	55	65
Bargellini et al[24]	22	RECIST1.1	0	1	5	16
		mRECIST	0	4	5	13
Ronot et al[14]	64	RECIST1.1	2		43	19
		mRECIST	18		29	17
Salvaggio et al[25]	17	RECIST1.1	0	2	10	5
		mRECIST	0	3	10	4
McNamara et al[26]	21	RECIST1.1	0	2	19	0
		mRECIST	1	6	14	0
Takada et al[15]	175	RECIST1.1	4	11	80	80
		mRECIST	5	20	72	78
Gavanier et al[27]	60	RECIST1.1	0	2	28	30
		mRECIST	0	4	27	29
Ikeda et al[28]	42	RECIST1.1	0	11	25	6
		mRECIST	0	17	19	6
Pelosof et al[29]	379	RECIST1.1	0	25	354	
		mRECIST	2	38	339	
Kaneko et al[30]	40	RECIST1.1	1	9	21	9
		mRECIST	3	12	9	4
Kawamura et al[31]	51	RECIST1.1	0	26	21	4
		mRECIST	6	32	9	4
Kuzuya et al[32]	10	RECIST1.1	0	0	8	2
		mRECIST	0	1	7	2
Maruta et al[33]	131	RECIST1.1	2	22	78	29
		mRECIST	3	59	42	27
Yamamichi et al[34]	22	RECIST1.1	1	1	12	8
		mRECIST	1	1	7	13
Murakami et al[19]	27	RECIST1.1	0	0	16	11
		mRECIST	1	2	13	11
He et al[35]	86	RECIST1.1	0	8	54	24

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		mRECIST	0	14	48	24
Nair et al[36]	LEN: 478	RECIST1.1	19		459	
		mRECIST	41		437	
	SOR: 476	RECIST1.1	7		469	
		mRECIST	12		464	
Salem et al[37]	158	RECIST1.1	18		140	
		mRECIST	22		136	
Yamashige et al[38]	11	RECIST1.1	6		5	
		mRECIST	9		2	

mRECIST, modified Response Evaluation Criteria in Solid Tumors; RECIST 1.1, Response Evaluation Criteria in Solid Tumors version 1.1; CR, complete response; PR, partial response; SD, stable disease; PD, progressive disease.

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Table 4. Inter-method agreement between RECIST 1.1 and mRECIST criteria.

Tumor response by RECIST1.1	Tumor response by mRECIST				Total
	CR	PR	SD	PD	
CR	6	0	0	0	6
PR	5	21	0	0	26
SD	4	41	168	5	215
PD	1	4	6	111	122
Total	16	66	174	116	372

RECIST 1.1, Response Evaluation Criteria in Solid Tumors version 1.1; mRECIST, modified Response Evaluation Criteria in Solid Tumors; CR, complete response; PR, partial response; SD, stable disease; PD, progressive disease.

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Figure 1. Flow diagram of study selection

Figure 2. Forest plot for HR for overall survival (responders vs non-responders) according to mRECIST

Figure 3. Forest plot for HR for overall survival (responders vs non-responders) according to RECIST1.1

Figure 4. Subgroup analysis for the association between response and overall survival according to mRECIST.

Figure 5. Subgroup analysis for the association between response and overall survival according to RECIST1.1.

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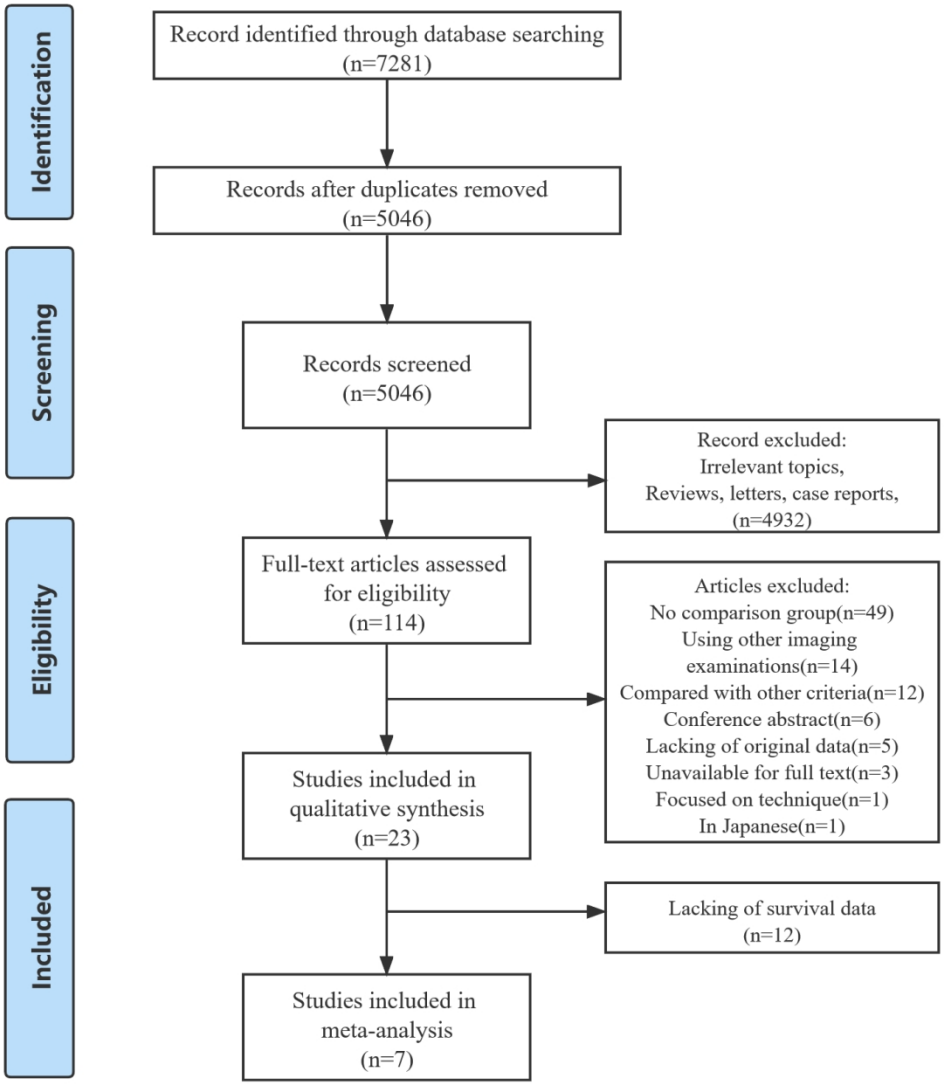


Fig 1

529x603mm (72 x 72 DPI)

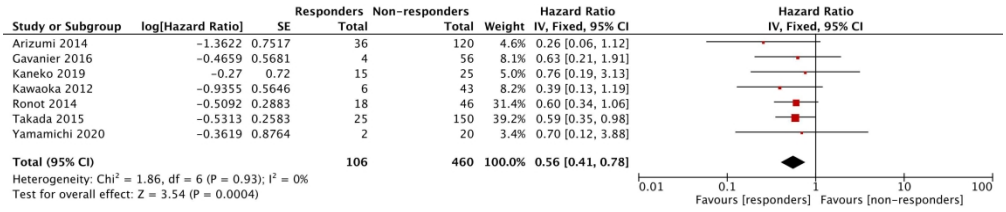

Responders Non-responders Hazard Ratio Hazard Ratio
Study or Subgroup log[Hazard Ratio] SE Total Total Weight IV, Fixed, 95% CI IV, Fixed, 95% CI
Arizumi 2014 -1.3622 0.7517 36 120 4.6% 0.26 [0.06, 1.12]
Gavanier 2016 -0.4659 0.5681 4 56 8.1% 0.63 [0.21, 1.91]
Kaneko 2019 -0.27 0.72 15 25 5.0% 0.76 [0.19, 3.13]
Kawaoka 2012 -0.9355 0.5646 6 43 8.2% 0.39 [0.13, 1.19]
Ronot 2014 -0.5092 0.2883 18 46 31.4% 0.60 [0.34, 1.06]
Takada 2015 -0.5313 0.2583 25 150 39.2% 0.59 [0.35, 0.98]
Yamamichi 2020 -0.3619 0.8764 2 20 3.4% 0.70 [0.12, 3.88]
Total (95% CI) 106 460 100.0% 0.56 [0.41, 0.78]
Heterogeneity: Chi² = 1.86, df = 6 (P = 0.93); I² = 0%
Test for overall effect: Z = 3.54 (P = 0.0004)
0.01 0.1 1 10 100
Favours [responders] Favours [non-responders]

Figure 2

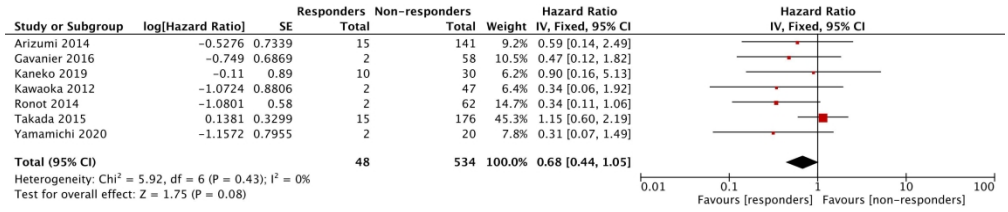

Responders
Non-responders
Hazard Ratio
Hazard Ratio
Study or Subgroup
log[Hazard Ratio]
SE
Total
Total
Weight
IV, Fixed, 95% CI
IV, Fixed, 95% CI
Arizumi 2014 -0.5276 0.7339 15 141 9.2% 0.59 [0.14, 2.49]
Gavanier 2016 -0.749 0.6869 2 58 10.5% 0.47 [0.12, 1.82]
Kaneko 2019 -0.11 0.89 10 30 6.2% 0.90 [0.16, 5.13]
Kawaoka 2012 -1.0724 0.8806 2 47 6.4% 0.34 [0.06, 1.92]
Ronot 2014 -1.0801 0.58 2 62 14.7% 0.34 [0.11, 1.06]
Takada 2015 0.1381 0.3299 15 176 45.3% 1.15 [0.60, 2.19]
Yamamichi 2020 -1.1572 0.7955 2 20 7.8% 0.31 [0.07, 1.49]
Total (95% CI) 48 534 100.0% 0.68 [0.44, 1.05]
Heterogeneity: Chi² = 5.92, df = 6 (P = 0.43); I² = 0%
Test for overall effect: Z = 1.75 (P = 0.08)
0.01
0.1
1
10
100
Favours [responders]
Favours [non-responders]

Figure 3

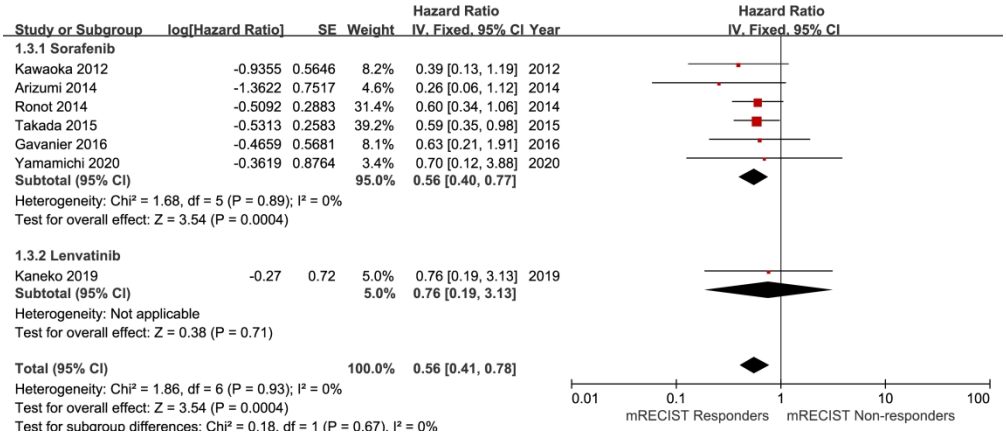


Fig 4

195x84mm (600 x 600 DPI)

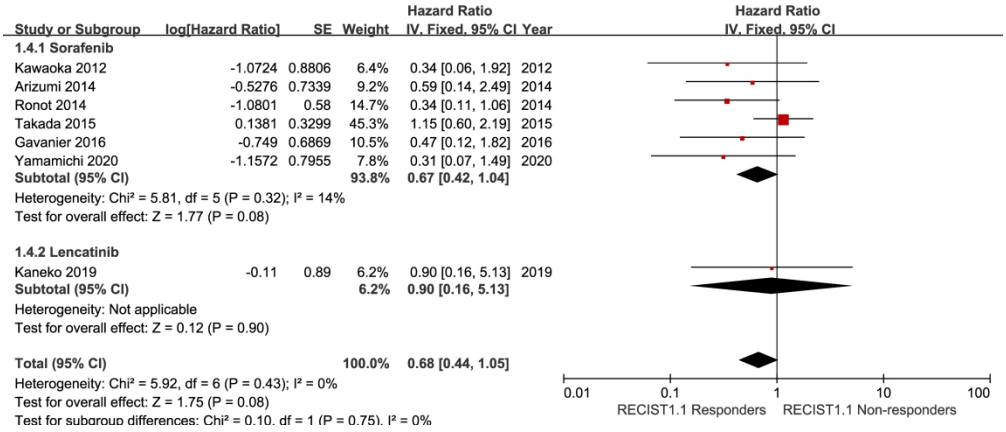
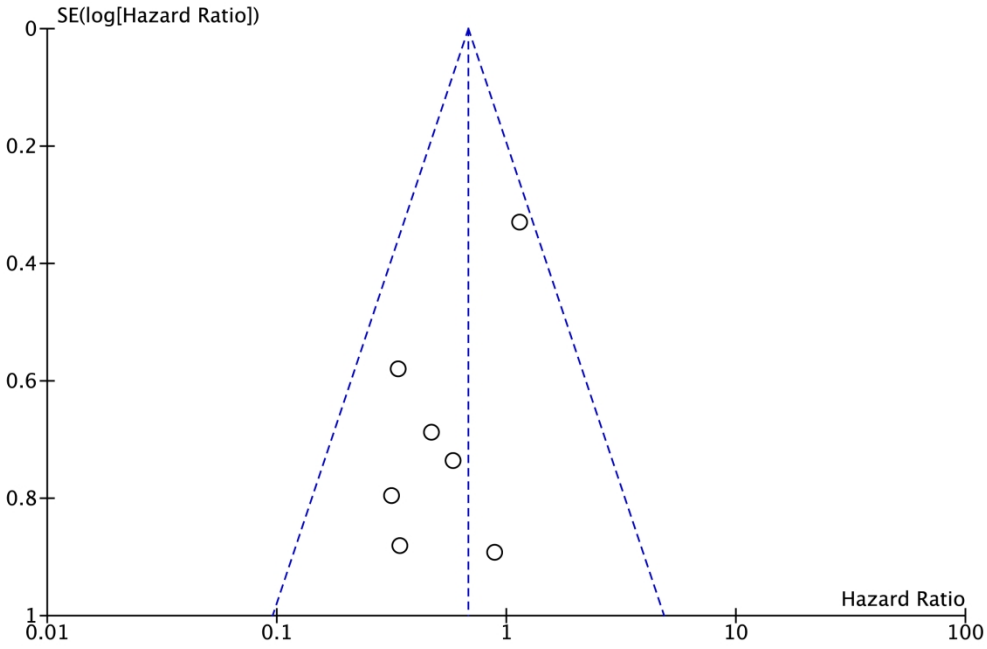
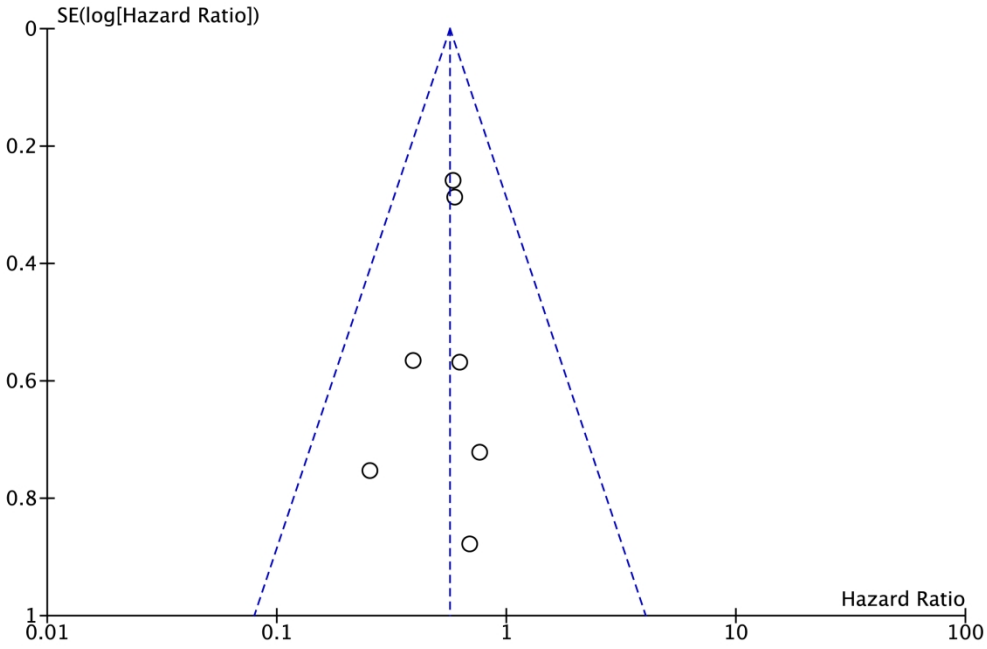


Fig 5

195x83mm (600 x 600 DPI)



211x141mm (300 x 300 DPI)



211x141mm (300 x 300 DPI)

Supplementary table1. Full search strategy for all databases

Database	Full search strategy
EMBASE	#1 'liver cell carcinoma'/exp #2 'liver neoplasms':ti,ab OR 'liver neoplasm':ti,ab OR 'hepatic neoplasms':ti,ab OR 'hepatic neoplasm':ti,ab OR 'cancer of liver':ti,ab OR 'hepatocellular cancer':ti,ab OR 'hepatocellular cancers':ti,ab OR 'hepatic cancer':ti,ab OR 'hepatic cancers':ti,ab OR 'liver cancer':ti,ab OR 'liver cancers':ti,ab OR 'cancer of the liver':ti,ab OR 'hepatocellular carcinomas':ti,ab OR 'adult liver cancer':ti,ab OR 'adult liver cancers':ti,ab OR 'liver cell carcinoma':ti,ab OR 'liver cell carcinomas':ti,ab OR 'hepatocellular carcinoma':ti,ab OR 'hepatoma':ti,ab OR 'hepatomas':ti,ab OR 'hepatic carcinoma':ti,ab OR hcc:ti,ab #3 #1 OR #2 #4 'response evaluation criteria in solid tumors'/exp #5 recist #6 #4 OR #5 #7 #3 AND #6
Pubmed	#1 ((((Hepatic carcinoma[Title/Abstract]) OR (HCC[Title/Abstract])) OR ((((((((((((((((((Carcinomas, Hepatocellular[Title/Abstract]) OR (Hepatocellular Carcinomas[Title/Abstract])) OR (Liver Cell Carcinoma, Adult[Title/Abstract])) OR (Liver Cancer, Adult[Title/Abstract])) OR (Adult Liver Cancer[Title/Abstract])) OR (Adult Liver Cancers[Title/Abstract])) OR (Cancer, Adult Liver[Title/Abstract])) OR (Cancers, Adult Liver[Title/Abstract])) OR (Liver Cancers, Adult[Title/Abstract])) OR (Liver Cell Carcinoma[Title/Abstract])) OR (Carcinoma, Liver Cell[Title/Abstract])) OR (Carcinomas, Liver Cell[Title/Abstract])) OR (Cell Carcinoma, Liver[Title/Abstract])) OR (Cell Carcinomas, Liver[Title/Abstract])) OR (Liver Cell Carcinomas[Title/Abstract])) OR (Hepatocellular Carcinoma[Title/Abstract])) OR (Hepatoma[Title/Abstract])) OR (Hepatomas[Title/Abstract]))) OR ((((((((((((((((((((Neoplasms, Hepatic[Title/Abstract]) OR (Neoplasms, Liver[Title/Abstract])) OR (Liver Neoplasm[Title/Abstract])) OR (Neoplasm, Liver[Title/Abstract])) OR (Hepatic Neoplasms[Title/Abstract])) OR (Hepatic Neoplasm[Title/Abstract])) OR (Neoplasm, Hepatic[Title/Abstract])) OR (Cancer of Liver[Title/Abstract])) OR (Hepatocellular Cancer[Title/Abstract])) OR (Cancers, Hepatocellular[Title/Abstract])) OR (Hepatocellular Cancers[Title/Abstract])) OR (Hepatic Cancer[Title/Abstract])) OR (Cancer, Hepatic[Title/Abstract])) OR (Cancers, Hepatic[Title/Abstract])) OR (Hepatic Cancers[Title/Abstract])) OR (Liver Cancer[Title/Abstract])) OR (Cancer, Liver[Title/Abstract])) OR (Cancers, Liver[Title/Abstract])) OR (Liver Cancers[Title/Abstract])) OR (Cancer of the Liver[Title/Abstract])) OR (Cancer, Hepatocellular[Title/Abstract]))) OR (("Liver Neoplasms"[Mesh]) OR "Carcinoma, Hepatocellular"[Mesh]) #2 (RECIST) OR ("Response Evaluation Criteria in Solid Tumors"[Mesh]) #3 #1 AND #2
Web of Science	#1TS=(LiverNeoplasms OR Neoplasms, Hepatic OR Neoplasms, Liver OR Liver Neoplasm OR Neoplasm, Liver OR Hepatic Neoplasms OR Hepatic Neoplasm OR Neoplasm, Hepatic OR Cancer of Liver OR Hepatocellular Cancer OR Cancers, Hepatocellular OR Hepatocellular Cancers OR Hepatic Cancer OR Cancer, Hepatic OR Cancers, Hepatic OR Hepatic Cancers OR Liver Cancer OR Cancer, Liver OR Cancers, Liver OR Liver Cancers OR Cancer of the L

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	iver OR Cancer, Hepatocellular OR Carcinoma, Hepatocellular OR Carcinomas, Hepatocellular OR Hepatocellular Carcinomas OR Liver Cell Carcinoma, Adult OR Liver Cancer, Adult OR Adult Liver Cancer OR Adult Liver Cancers OR Cancer, Adult Liver OR Cancers, Adult Liver OR Liver Cancers, Adult OR Liver Cell Carcinoma OR Carcinoma, Liver Cell OR Carcinomas, Liver Cell OR Cell Carcinoma, Liver OR Cell Carcinomas, Liver OR Liver Cell Carcinomas OR Hepatocellular Carcinoma OR Hepatoma OR Hepatomas OR Hepatic carcinoma OR HCC) #2TS=(Response Evaluation Criteria in Solid Tumors OR RECIST) #3 #1AND #2
Cochrane Library	#1 MeSH descriptor: [Liver Neoplasms] explode all trees #2 (Liver Neoplasms):ti,ab,kw OR (Liver Neoplasm):ti,ab,kw OR (Hepatic Neoplasms):ti,ab,kw OR (Hepatic Neoplasm):ti,ab,kw OR (Cancer of Liver):ti,ab,kw #3 (Hepatocellular Cancer):ti,ab,kw OR (Hepatocellular Cancers):ti,ab,kw OR (Hepatic Cancer):ti,ab,kw OR (Liver Cancer):ti,ab,kw OR (Liver Cancers):ti,ab,kw #4 (Cancer of the Liver):ti,ab,kw OR (Hepatocellular Carcinomas):ti,ab,kw OR (Liver Cell Carcinoma, Adult):ti,ab,kw OR (Adult Liver Cancer):ti,ab,kw OR (Adult Liver Cancers):ti,ab,kw #5 (Liver Cell Carcinoma):ti,ab,kw OR (Liver Cell Carcinomas):ti,ab,kw OR (Hepatocellular Carcinoma):ti,ab,kw OR (Hepatoma):ti,ab,kw OR (Hepatomas):ti,ab,kw #6 (Hepatic carcinoma):ti,ab,kw OR (HCC):ti,ab,kw #7 #1 or #2 or #3 or #4 or #5 or #6 #8 MeSH descriptor: [Response Evaluation Criteria in Solid Tumors] explode all trees #9 (RECIST) #10 #8 or #9 #11 #7 and #10

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	#1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	#2; #3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	#5; #6
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	#6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	#7; #8
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	#7
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	#7; Table S1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	#7; #8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	#8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	#8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	#8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	#9; #10
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	#9; #10
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	#9; #10
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	#9; #10
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	#9; #10
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	#9; #10
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	#9; #10
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	#9; #10
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	#9; #10
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	#10; #11
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	#10; #11; Fig 1
Study characteristics	17	Cite each included study and present its characteristics.	#11, Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	#12
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	#12; #13; #14
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	#13; #14
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	#13; #14
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	#14
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	#14
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	#14
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	#14; #15
	23b	Discuss any limitations of the evidence included in the review.	#18
	23c	Discuss any limitations of the review processes used.	#18
	23d	Discuss implications of the results for practice, policy, and future research.	#16; #17; #18
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	#3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	#2
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	#1
Competing interests	26	Declare any competing interests of review authors.	#1
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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